

1. Administrative & Study Identification

Full Study Title: Real World-clinical Outcomes of Tucatinib, Trastuzumab, and Capecitabine Following Trastuzumab Deruxtecan (T-DXd) for the Treatment of HER2-positive Metastatic Breast Cancer. **Short Title/Acronym:** rw-Outcomes of Tucatinib Triplet Post-T-DXd **Protocol Number:** C4251025 **NCT Number:** NCT06961331 **Sponsor Name:** Pfizer **Investigational Product:** Tucatinib, Trastuzumab, and Capecitabine (Tucatinib Triplet Therapy) **Phase of Development:** Not Applicable **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To describe 3L and 4L real-world (rw) outcomes of the tucatinib-trastuzumab-capecitabine triplet therapy immediately following Trastuzumab Deruxtecan (T-DXd) therapy in patients diagnosed with HER2+ metastatic breast cancer (mBC) in the United States. **Secondary Objectives:** To evaluate time to next treatment, time to treatment discontinuation, overall survival (OS), and real-world progression-free survival (rwPFS).

2.2 Background & Rationale To address the gap in real-world clinical evidence regarding optimal sequencing and effectiveness of the tucatinib-trastuzumab-capecitabine triplet regimen immediately following disease progression on T-DXd in patients with HER2-positive metastatic breast cancer.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Non-interventional, Retrospective cohort analysis) **Control Type:** Single-arm **Blinding:** Open-label (De-identified retrospective data) **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 93 **Number of Sites:** United States (Flatiron Health Database) **Estimated Study Start:** 07-May-2025 **Estimated Primary Completion:** 25-Sep-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. mBC diagnosis between January 1, 2017 and 6 months prior to database cutoff. 2. Evidence of human

epidermal growth factor receptor 2 (HER2) receptor positivity prior to or up to 90 days following the mBC diagnosis date. 3. At least 18 years old on the mBC diagnosis date. 4. Treated with systemic anticancer treatment in the metastatic setting, i.e., post mBC diagnosis. 5. Treated with T-DXd in the 2L or 3L setting, followed immediately by the tucatinib triplet, and at least 1 additional prior HER2-targeted regimen.

4.2 Exclusion Criteria 1. Patients with evidence of other cancers prior to the mBC diagnosis date will be identified with the following International Classification of Diseases, 9th Revision, Clinical Modification (ICD-9-CM) and International Classification of Diseases, 10th Revision, Clinical Modification (ICD-10-CM) codes in all databases: 140.xx - 195.xx, 200.xx - 208.xx, C00.xx - C76.xx, C81.xx - C96.xx. (exclude - ICD-9: 174.X, 175.X; ICD-10: C50.XX). 2. Patients treated with tucatinib or T-DXd prior to the 2L or 3L T-DXd treatment. 3. Patients participating in clinical trials. 4. Patients receiving concomitant endocrine therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Standard-of-care real-world dosing for the Tucatinib-Trastuzumab-Capecitabine triplet. **Route of Administration:** Oral (Tucatinib, Capecitabine) and Intravenous/Subcutaneous (Trastuzumab). **Duration of Treatment:** Until treatment discontinuation for any reason, progression, or death.

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care medications as prescribed in real-world practice. **Prohibited:** Concomitant endocrine therapy.

6. Study Timeline & Visit Schedule

(Note: As this is a retrospective database analysis, standard visit windows are replaced by data abstraction milestones.) |
Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | N/A | Flatiron Database query for mBC diagnosis and HER2+ status between Jan 2017 and 6-months pre-cutoff | | **Baseline** | Index Date | Initiation of the Tucatinib-Trastuzumab-Capecitabine triplet immediately post-T-DXd | | **Interim** | Real-world follow-up | Abstraction of treatment duration, disease progression, and sequential therapy lines | | **End of Study** | Database Cutoff | Final evaluation of Primary Outcomes |

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Time to Next Treatment**: Defined as the time in months from the initiation of therapy to the initiation of subsequent LOT or death. 2. **Time to Discontinuation**: Defined as the time in months from initiation of therapy to discontinuation for any reason or death. 3. **Overall survival (OS)**: Defined as the time in months from initiation of therapy to death from any cause.

7.2 Secondary & Exploratory Endpoints - Real-world Progression-Free Survival (rwPFS): Time in months from initiation of therapy to progression or death from any cause.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: N/A - This is a non-interventional, retrospective analysis using de-identified electronic health records. **Serious Adverse Events (SAEs)**: N/A. **Data Monitoring Committee (DMC)**: N/A.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Patients in the US Flatiron Dataset diagnosed with HER2+ mBC treated with the tucatinib triplet post-T-DXd setting. **Sample Size Justification**: Exploratory real-world data cohort based on an estimated 93 patients satisfying the eligibility criteria within the Flatiron database. **Handling of Missing Data**: Standard handling/imputation parameters established for retrospective Flatiron electronic health record abstractions.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and local patient privacy laws protecting de-identified data. **Monitoring Plan**: Programmatic data validation and source documentation abstraction checks specific to the Flatiron Health Database. **Audit Trail**: Secure maintenance of de-identified patient dataset queries and analytical audit logs.

11. Study Identifiers & Governance

Primary ID: {{C4251025}} Registry Number: {{NCT06961331}} _
Sponsor/Lead Organization: {{Pfizer}} Study Title: {{Real World-clinical Outcomes of Tucatinib, Trastuzumab, and Capecitabine Following Trastuzumab Deruxtecan (T-DXd) for the Treatment of HER2-positive Metastatic Breast Cancer.}}

Title: {{Real-world Clinical Outcomes of Tucatinib, Trastuzumab, and Capecitabine Following Trastuzumab Deruxtecan for the Treatment of HER2-postive Metastatic Breast Cancer}}

12. Clinical Framework (Adapted for HER2+ mBC)

Primary Condition: {{Metastatic HER2+ Advanced Breast Cancer}}
Diagnosis Threshold: {{Evidence of HER2 receptor positivity prior to or up to 90 days following mBC diagnosis}} Medical
Methodology: {{Real-world standard-of-care utilizing Flatiron Health Database}} Optimization Target: {{Triplet therapy sequencing immediately following T-DXd}}

13. Study Procedures & Methodology

Management Pathway: {{Retrospective evaluation of 3L and 4L real-world outcomes}} Education Protocol (HCP): {{N/A - Non-interventional retrospective study}} Education Protocol (Patient): {{N/A - Non-interventional retrospective study}}
Quality Audit Mechanism: {{De-identified patient data verification to preserve privacy via Flatiron Health Database}}

14. Observation & Follow-Up Schedule

Total Duration: {{Variable; measured from initiation of therapy to progression, subsequent LOT, database cutoff, or death}}
Onsite Visit Cadence: {{N/A - Electronic Health Record Abstraction}} Remote Monitoring Frequency: {{N/A}} Checklist Review Interval: {{N/A}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				